CLAIMS

What is claimed is:

1. A central processing unit, comprising:

- a plurality of parametric registers, wherein each parametric register is configured to store a data structure representing a mathematical expression of a number; and
- a Parametric Logic Unit (PLU) communicatively coupled to the parametric registers, the PLU configured to execute instructions by performing symbolic mathematical operations on the data structures stored in said registers.

2. The CPU of claim 1, wherein the data structure represents a number as a sequence of operands and operators.

3. The CPU of claim 1, wherein the data structure represents a mathematical constant as a native symbol.

4. The CPU of claim 3, wherein the mathematical constant is Pi or Euler's number.

5. The CPU of claim 1, wherein the PLU, in executing an arithmetic instruction on a first and a second parametric register, is configured to generate a new data structure representing the symbolic combination of the expressions from the first and second registers.

6. The CPU of claim 1, further comprising an instruction decoder configured to interpret a specialized instruction set architecture including instructions for loading, storing, and symbolically manipulating the data structures in the parametric registers.

7. The CPU of claim 6, wherein the instruction set includes an evaluation instruction that causes the PLU to compute a numerical value from a data structure to a specified precision.

8. A method for computation within a central processing unit, comprising:

- storing a first number as a first data structure representing a first mathematical expression in a first parametric register;
- storing a second number as a second data structure representing a second mathematical expression in a second parametric register; and
- executing an instruction by creating a third data structure representing the symbolic mathematical combination of the first and second expressions, and storing said third data structure in a third parametric register.

9. The method of claim 8, wherein the execution of the instruction occurs without evaluating the first and second mathematical expressions to fixed-precision numerical values.